THE OHIO STATE UNIVERSITY
DEPARTMENT OF ENGINEERING EDUCATION

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Exploring Perspectives on Pre-College Robotics Education: A Social Media–Based *Topic* and *Emotion* Analysis

Work-in-Progress · 4,212 Reddit posts (2023–2025) · BERTopic-style modeling + LLM-based emotion analysis

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01 INTRODUCTION

Pre-college institutions increasingly use robotics to teach STEM concepts and spark interest in engineering, often through competitions such as FIRST LEGO League and FIRST Tech Challenge. Teachers are central to facilitating these experiences.

Yet teachers face real challenges: adding content under standardized-testing pressure and limited technical knowledge of robotics, which can lower their self-efficacy and willingness to teach it.

To navigate this, pre-college teachers turn to online forums—Reddit subreddits and Facebook groups—to share experiences, ask questions, and find support. These discussions are a rich, under-examined window into their experiences.

02 WHY SOCIAL MEDIA & NLP

Educational research increasingly treats online networks as learning spaces and mines them with Natural Language Processing. Prior work used BERT-based topic modeling to study ChatGPT discourse [5] and LDA to map discussions in higher-education subreddits [6].

This study builds on that work to surface broad, large-scale patterns beyond context-specific case studies—pairing data-driven topic modeling with fine-grained, LLM-based emotion analysis to map both *what* is discussed and *how it feels*.

03 RESEARCH QUESTION

> *What themes characterize online discussions about robotics education, and how are these themes associated with distinct emotional profiles?*

04 DATA & METHODS

Source: Reddit — five robotics-competition communities (r/FLL, r/FTC, r/FRC, r/vex, r/lego), collected via PRAW using a matrix of **14 education keywords** (course, curriculum, teach, coach, student, K–12, programming, competition…).

Window: Aug 1, 2023 – Sep 24, 2025. After deduplication (by post ID) and outlier screening: 4,212 unique posts (titles, body, timestamps).

Topic modeling: each post was embedded with OpenAI *text-embedding-3-small* (1536-dim), clustered with HDBSCAN, merged into coherent themes, and labeled by GPT-4o-mini. **Emotion analysis:** an LLM *gpt_emotions* function scored nine emotions (0–1) over a stratified sample (≤30 posts/topic), keeping each topic's top five.

ANALYTICAL PIPELINE

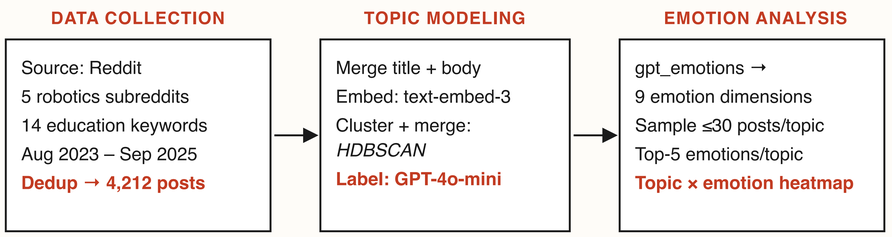


FIG 1. Three-phase analytical workflow: Reddit collection → topic modeling → emotion analysis.

BY THE NUMBERS

4,212	5	10	9
REDDIT POSTS	SUBREDDITS	THEMES	EMOTIONS

05 RESULTS: WHAT IS DISCUSSED

Topic modeling surfaced **nine substantive themes** plus one noise cluster. The distribution is highly uneven—a few core concerns dominate the conversation.

TOPIC (N = 4,212)	POSTS	%
Exploring Robotics Ed. Resources	2,323	55.2
Seeking Mentorship	330	7.8
Troubleshooting Programming	299	7.1
Enchanted Class Concepts	18	0.4
Sharing Resources	8	0.2
Challenges in Online Resources	6	0.1
Collecting Robotics Pins	6	0.1
Future Tools for Robotics	6	0.1
Dean's List & Involvement	5	0.1
Noise (no coherent cluster)	*1,213*	*28.8*

The top three topics—accessing resources, mentorship, and troubleshooting—account for ~70% of substantive posts, signaling that practical, support-seeking concerns dominate online robotics-education discussion.

KEY EMOTIONAL PATTERNS

Exploration drives positive affect. Resource-, concept-, and future-tool topics are dominated by curiosity, excitement, and joy—learning- and discovery-oriented engagement.

Technical & access friction breeds negativity. "Challenges in Online Resources" peaks in frustration (0.73) and disappointment; troubleshooting pairs curiosity with frustration and anxiety.

Mixed profiles reveal tension. Mentorship blends curiosity with anxiety over roles; "Dean's List" mixes pride with institutional pressure.

06 RESULTS: HOW IT FEELS

Topics skew toward curiosity and excitement, with frustration and anxiety concentrated in technical and access-related themes.

TOPIC × EMOTION INTENSITY (0–1)

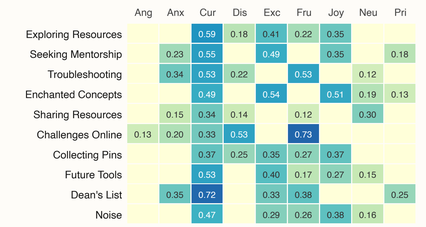


FIG 2. Mean emotion intensity per topic (top-5 emotions retained; LLM-scored).

07 CONCLUSION & FUTURE WORK

Online robotics-education discussion is dominated by resource-seeking and colored by curiosity and excitement, with frustration tied to programming and platform-access issues. Future work will refine post classification, add hierarchical topic modeling to reduce the 28.8% noise, and use human coding to separate teacher, student, and parent voices—centering teacher perspectives.

SELECTED REFERENCES

[1] S. Anwar et al., "A systematic review of studies on educational robotics," *J. Pre-Coll. Eng. Educ. Res.*, 9(2), 2019.
[3] C. A. Berry et al., "Robotics for all ages: A standard robotics curriculum for K-16," *IEEE Robot. Autom. Mag.*, 23(2), 2016.
[5] L. Li et al., "ChatGPT in education: a discourse analysis of worries on social media," *Educ. Inf. Technol.*, 29(9), 2024.
[6] P. S. Muljana et al., "Adjusting sails for changing winds: exploring Reddit use for professional purposes in higher education," *J. Comput. High. Educ.*, 34(3), 2022.

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